

DPhil Thesis Draft



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List of Abbreviations

- 1-D, 2-D** . . . One- or two-dimensional, referring in this thesis to spatial dimensions in an image.
- Otter** One of the finest of water mammals.
- Hedgehog** . . . Quite a nice prickly friend.

1

Introduction

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1.1 Motivation

Motivation

1.2 Carbon-based Field Effect Transistor

Detail here the carbon nanotube journey.

<i>Device type</i>	$I_{\text{on}}/I_{\text{off}}$	SS (mV/dec)	g_{mt} ($\mu\text{S cm}^{-1}$)	L_{ch} (nm)	f_T [f_{max}] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[1]
	10^5	~ 85	40	15	—	[2]
	10^5	~ 85	20	20	—	[2]
	10^5	60	20	200	—	[3]
	10^5	~ 63	—	200	—	[4]
	10^4	80	50	240	—	[5]
	—	—	143	800	[1]	[6]
	10^4	—	—	1,500	1.5 – 3.0	[7]
	—	70	$12^\dagger$	2,000 ¹	—	[8]
	10^8	—	—	$< 10,000^1$	—	[9]
	—	—	- 500	—	30	[10]
Top gate	10^6	—	20^2	80	—	[11]
	2	—	~ 50	100	25	[12]
	50	—	310	100	~ 70 [80]	[13]
	—	—	0.3	210	—	[14]
	10^5	—	33.5	85,000	—	[15]

Table 1.1: Figures of merit for a few CNFET reported in literature. S : subthreshold swing; g_m : intrinsic transconductances; L_{ch} : channel length; μ : carriers mobility

[†]Normalized by the tube width.

¹ Referred to the gated length of a nanotube.

² For the n-type transistor, whereas 10 μS were measured for the p-type.

<i>Device type</i>	I_{on} (A)	I_{off} (A)	$I_{\text{on}}/I_{\text{off}}$	g_{mt} (μS)	SS (mV/dec)
Conventional	1.41×10^{-5}	4.06×10^{-10}	3.4×10^4	29	99.6
Partial S/D-doped	1.50×10^{-5}	2.33×10^{-8}	6.4×10^2	63.6	93.1
Schottky-Barrier	3.26×10^{-6}	2.31×10^{-8}	1.41×10^2	19.6	109.8
Tunnel	5.53×10^{-6}	1.16×10^{-8}	4.76×10^2	13	107.9

Table 1.2: Comparative simulation for the four major type of CNFETs as reported in [16]. These results are obtained by setting $V_{\text{sd}} = 0.5$ V.

1.3 Graphene Nanoribbons

GNR.

It will be interesting to discuss the $E_g \propto 1/W$ relation as the model used for describing gnr is the particle in the box, but in the toy model (particle in a box) as a relation of the kind $E_g \propto 1/W^2$.

1.4 Delocalised π -electron Systems

Porphyrin

2

Foundations

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2.1 Charge Transport In Zero-Dimensional Objects

quantum dots transport physics

2.2 Charge Transport in One-Dimensional Objects

nanowires transport physics

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

What are the necessary changes that one has to take into account when dealing also with other extensive variable such as a temperature gradient

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

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Instrumentations

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3.1 Room Temperature DC Measurements

Describe here the probe station setup]

3.2 Dipstick DC Measurements

Describe here the dipstick used for both liquid N₂ and liquid He

3.3 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

3.4 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it usepackage

3.5 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

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Fabrication

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4.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

4.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and lithographic patterning of graphene nanogap

4.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

4.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

4.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

4.3.3 Thermoelectric Device

Common and different features of the thermoelectric

4.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techniques How this has been qualitatevely addressed.

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Room Temperature Graphene Nanoribbon Single-Electron Transistor

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5.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)
Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

5.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

5.4 Summary

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Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

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6.1 Introduction

A brief intro on how why it is important to study this

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks i s intrinsically scarce in
molecular electronics.

6.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

6.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

6.5 Summary

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Thermocurrent Spectroscopy of covalently Functionalized Graphene Nanoribbons

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7.1 Introduction

A brief intro on how why it is important to study this what has been done.

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks is intrinsically scarce in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

7.4 Summary

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Transport Spectroscopy of butadiyne-linked Dy–Tb Double-Porphyrin Unit

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8.1 Introduction

A brief intro on how why it is important to study this

8.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters)

Explained why the scarcity of characterisation checks i s intrinsically scarce in molecular electronics.

8.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest bandgap distance.

8.4 Summary

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Conclusions and Outlook

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9.1 Conclusions

Here the final conclusions

9.2 Outlook

Appendices

Cor animalium, fundamentum est vitæ, princeps omnium, Microcofimi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [17]



Review of Cardiac Physiology and Electrophysiology

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Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

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A.2 Mechanical Cycle

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A.3 Electrical Cycle

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A.4 Cellular Electromechanical Coupling

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*The first kind of intellectual and artistic personality
belongs to the hedgehogs, the second to the foxes ...*

— Sir Isaiah Berlin [18]

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